

M.S. NAZ HIGH SCHOOL

Excellence in Matriculation & Academic Leadership | Est. 1989 | BISE Gujranwala Code: 112199

Single National Curriculum (SNC) & PECTAA Standards Aligned - Session 2026-2027

CLASS 10 (10TH MATRIC) - MATHEMATICS (SCIENCE GROUP)

Official Syllabus, Board Paper Scheme & High-Yield Examination Revision Notes

Group: Science Group | Total Marks: 75 Marks | Time: 2 Hours 30 Minutes

SECTION 1: BISE GUJRWANWALA EXAMINATION BLUEPRINT & WEIGHTAGE

- Section A: Objective (MCQs):

15 Compulsory MCQs (15 Marks).

- Section B: Short Questions:

Attempt 18 out of 27 Short Questions (Q2, Q3, Q4: 6/9 each = 36 Marks).

- Section C: Long Questions:

Attempt 3 out of 5 Long Questions (8 Marks each = 24 Marks). Question 9 (Circle Theorems from Ch 9 or Ch 12) is COMPULSORY.

SECTION 2: CHAPTER-WISE HIGH-YIELD REVISION NOTES & CORE SLOs

Chapter 1: Quadratic Equations:

Standard form ($ax^2 + bx + c = 0$, $a \neq 0$), Pure quadratic ($b = 0$), Solution methods: 1. Factorization, 2. Completing the square, 3. Quadratic Formula ($x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$), Reducible equations (reciprocal, exponential, radical equations).

Chapter 2: Theory of Quadratic Equations:

Discriminant ($b^2 - 4ac$) and nature of roots (real, rational, irrational, imaginary), Cube roots of unity (1, ω , ω^2), Properties: $1 + \omega + \omega^2 = 0$, $\omega^3 = 1$, Sum ($S = -b/a$) and Product ($P = c/a$) of roots, Forming equation: $x^2 - Sx + P = 0$, Synthetic Division.

Chapter 3: Variations:

Ratio and Proportion, Direct Variation ($y = kx$), Inverse Variation ($y = k/x$), Joint Variation, Theorems: Invertendo, Alternando, Componendo, Dividendo, Componendo-Dividendo Theorem ($a/b = c/d \Rightarrow (a+b)/(a-b) = (c+d)/(c-d)$), K-method.

Chapter 4: Partial Fractions:

Proper vs Improper fractions, Resolution into partial fractions: Case I: Non-repeated linear factors, Case II: Repeated linear factors, Case III: Non-repeated irreducible quadratic factors.

Chapter 5: Sets and Functions:

Operations (Union, Intersection, Difference, Complement), De Morgan Laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$, Venn diagrams, Binary Relations, Domain and Range, Functions: Into, Onto, One-to-One, Bijective.

Chapter 6: Basic Statistics:

Frequency distribution, Measures of Central Tendency: Arithmetic Mean (Direct, Indirect methods), Geometric Mean, Harmonic Mean, Median, Mode, Measures of Dispersion: Range, Variance, Standard Deviation ($S = \sqrt{\sum(x - \text{mean})^2 / n}$).

Chapter 7: Introduction to Trigonometry:

Measurement of angles (Degree/Minute/Second vs Radians: $\pi \text{ rad} = 180^\circ$), Arc length ($l = r\theta$), Sector area ($A = \frac{1}{2} r^2 \theta$), Trigonometric ratios (sin, cos, tan, csc, sec, cot), Fundamental identities: $\sin^2 \theta + \cos^2 \theta = 1$; $1 + \tan^2 \theta = \sec^2 \theta$; $1 + \cot^2 \theta = \csc^2 \theta$, Heights and distances (elevation & depression).

Chapter 9: Chords of a Circle (Theorems):

Theorem 9.1: One and only one circle can pass through three non-collinear points. Theorem 9.2: A straight line drawn from the centre of a circle to bisect a chord is perpendicular to the chord. Theorem 9.4: Congruent chords of a circle are equidistant from the centre.

SECTION 3: MSNS SENIOR FACULTY BOARD EXAMINATION STRATEGY

- * Question 9 Circle Theorem is COMPULSORY (8 marks): Always draw the circle with a compass, draw given chord with a ruler, and label center O.
- * Show all calculation steps clearly when using Componendo-Dividendo theorem in variations.